**Mathematics (3)**
**(MTH 203,MTHN103,MTHn103)**
**Revision sheet**
For Mid-Term Examination
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# Differential Equations

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## First order ordinary differential equations

(1) $e^{y^2}\ dx + x^2 y\ dy = 0$

(2) $(4+x)\ y' = y^3$

(3) $e^{2y}\ y' = e^{x} + e^{x+2y}$

(4) $2xy\ y' = 1 + y^2$

(5) $ye^{-y}\ y' = \sin x$

(6) $\left(x^2 - xy + y^2\right)dx - xy\ dy = 0$

(7) $y' = \dfrac{2xy}{3x^2 - y^2}$

(8) $y' = \dfrac{1-2y-4x}{1+y+2x}$

(9) $ye^{x}\ dx + \left[e^{x} + (1+y)e^{y}\right]dy = 0$

(10) $\left(\cot y + x^2\right)dx = x\csc^2 y\ \ dy$

(11) $\left(1 + y^2 + xy^2\right)dx + y\left(x^2 + 2x + 1\right)dy = 0$

(12) $\left(2xy - \tan y\right)dx + \left(x^2 - x\sec^2 y\right)dy = 0$

(13) $\left(3xy - 4y - 1\right)dx + x\left(x-2\right)\ dy = 0$

(14) $\left(2xy\right)dx + \left(4y + 3x^2\right)\ dy = 0$

(15) $x\,y' - 2y = x^3 e^x$

(16) $y' = x - 2y\,\cot(2x)$

(17) $y' = y - xy^3 e^{-2x}$

(18) $x\,y' - y = x^3 y^4$

**Second and higher order ordinary differential equations**

(1) $y'' - 2y' - 8y = 0$

(2) $y'' + 10y' + 25y = 0$

(3) $y'' + 3y' + 6y = 0$

(4) $y^{(4)} - y = 0$

(5) $y''' - 4y'' + 5y' - 2y = 0$

(6) $y'' - 3y' + 2y = e^x$ $\qquad y = e^{2x}$

(7) $y'' - y' - 2y = 6x + 6e^x$

(8) $y'' - 4y' + 3y = 20\cos x$

(9) $y'' - 4y = 2\cos^2 x.$

(10) $y'' + 2y' + y = xe^{2x}.$

(11) $y'' + y = 3\sin 3x \cos 3x.$

(12) $y'' + y' = 2x + 3e^x.$

(13) $y'' + y = \cos x \tan x$